AberdeenGroup
A Harte-Hanks Company

Measuring the Returns from a Desktop Virtualization Program

September 2011

Dick Csaplar

Analyst Insight

Aberdeen Group
A Harte-Hanks Company

September, 2011

Measuring the Returns from a Desktop Virtualization Program

In May, 2011Aberdeen conducted a survey on the deployment of desktop virtualization, exploring the challenges faced, strategies employed, and ultimately the tangible business and operational benefits organizations have achieved as a result of this technology. Representatives from a diverse group of 76 organizations responded to the Aberdeen desktop virtualization survey.

This Analyst Insight is a detailed examination of how desktop performance varies between companies with desktop virtualization and those that have not deployed that technology. By examining both types of companies, Aberdeen created seven metrics of desktop performance that apply to both virtualized and non-virtualized desktop environments. These metrics can help organizations determine if they are getting the best possible results from their desktop virtualization program. Desktop virtualization directly and visibly touches each member of the program and affects their daily productivity.

These metrics are:

- Change in time for routine desktop maintenance
- Change in desktop support spending over the last 12 months
- Number of desktop downtime incidents
- Change in desktop downtime over the last 12 months
- The average time to remediate each downtime event
- The amount of data lost in each downtime event
- Employee satisfaction with their desktops

In addition Aberdeen presents several business practices and technology enablers used by successful companies that contributed to their greater desktop performance.

As this document will demonstrate, a virtualization program will improve the operational management of end-user desktops. Supporting a desktop virtualization program with the proper business practices and technology enablers will contribute to a successful desktop virtualization program.

Desktop Virtualization Program Overview

Aberdeen examined the companies that have deployed desktop virtualization and their experience with the technology. The responding organizations were fairly evenly spread with 38% of respondents from small

Analyst Insight

Aberdeen's Insights provide the analyst perspective of the research as drawn from an aggregated view of the research surveys, interviews, and data analysis

Definitions

Terms used in this report include:

√ Desktop virtualization - a concept that separates the desktop hardware and the applications, allowing IT to manage desktop devices as a group and not individually

√ Virtual Desktop Infrastructure (VDI) - a type of desktop virtualization that includes the servers and storage required to support the desktop computing devices

√ Thin client device - desktop device that requires another computer to fulfill most of its computing responsibilities, and includes some software / firmware

√ Zero client device - an extreme thin client device, possessing no software or firmware that requires management or updates businesses, (less than $50M in annual revenue), 32% from mid-sized and 30% from large enterprises (over $1B in yearly sales).

Table 1: How long have you had a desktop virtualization program?

	Overall	Small Enterprise	Mid-Sized Enterprise	Large Enterprise
More than 2 years	29%	15%	33%	45%
1 to 2 years	19%	15%	19%	25%
Less than 1 year	12%	12%	10%	10%

Source: Aberdeen Group, May 2011

Almost half (45%) of large organizations have had some form of desktop virtualization program for over two years, compared to only 15% for small enterprises. Likewise, 80% of large companies have already deployed desktop virtualization in at least some part of their company. Desktop virtualization, more so than server or storage virtualization, has been adopted mainly by larger organizations, with the market yet to see much penetration into small or mid-sized companies.

Table 2: When will you deploy a desktop virtualization program?

	Overall	Small Enterprise	Mid-Sized Enterprise	Large Enterprise
Will begin this year	15%	31%	0%	10%
Project planned for beyond this year	12%	4%	24%	10%
No plans	13%	23%	14%	0%

Source: Aberdeen Group, May 2011

"Study the hard and soft benefits carefully. Most probably you will face challenges from some unsatisfied end users."

~ IT Manager, Mid-Sized Investment Company, Middle

Thirty-one percent (31%) of small companies report that they will begin a desktop virtualization program this year with 24% of mid-sized organizations reporting they will soon begin deployment of a solution already in the planning phase. However, the primary adopters of desktop virtualization technology remain large enterprises. Twenty-three percent (23%) of small companies and 14% of mid-sized organizations report that they have no plans to deploy desktop virtualization at all. Clearly large enterprises, with their large and diverse desktop communities believe they stand to gain the most from deploying desktop virtualization.

Progress to Date

As with all new technologies, Aberdeen asked the survey respondents to report how widely they have deploying desktop virtualization to their end-users. Companies that report having a desktop virtualization program indicate that they have virtualized about one third, or 34%, of their desktops to date. They also report that when all current programs are complete they will have virtualized 61% of available desktops, or about twice as many as are virtualized today.

Table 3: How far are you in deploying desktop virtualization?

	Companies with a Desktop Virtualization Program
Average Percent of Desktops Virtualized	34%
Percent When all Projects Completed	61%

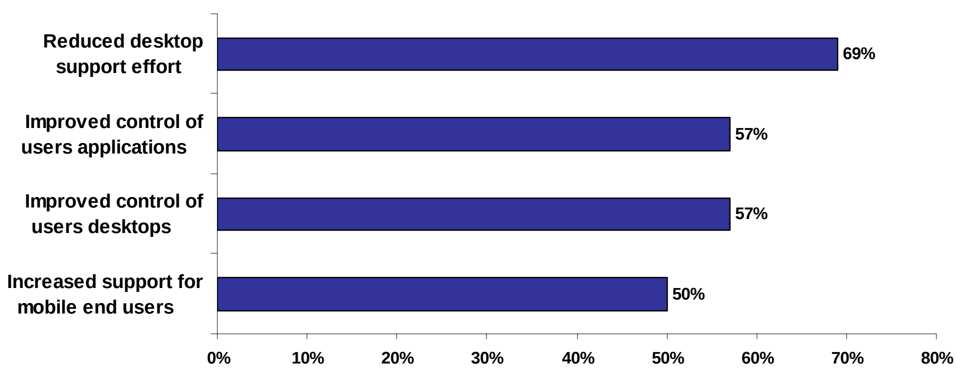
Source: Aberdeen Group, May 2011

Ten percent (10%) of respondents reported that they have already virtualized over 90% of their desktops. All of these respondents, however, were small operations with fewer than 100 employees. This corresponds to certain desktop virtualization technologies that allow a loose, one-size-fits-all implementation with limited end-user customization for a limited number of applications. Companies that could benefit from this type of solution should consider the technology options in the marketplace.

Virtualization Benefits

Aberdeen research shows what type of benefits organizations are getting as a result of their active desktop virtualization program.

Figure 1: Benefits of Desktop Virtualization Program



Source: Aberdeen Group, May 2011

"[When it comes to desktop virtualization] look for compliance and information security improvements instead of cost reductions as the ROI."

~ IT Manager, Large Aerospace Corporation, US

For most technologies that Aberdeen reviews, the deployment benefits comprise both operational and financial returns. With desktop virtualization this is very different. The top four benefits cited by survey respondents are almost exclusively operational.

The primary benefit of virtualization reported by almost 7 of 10 companies (69%) is being able to reduce the amount of effort required to support their desktop infrastructure. While reducing the need for desktop support resources could produce financial savings in the long run, companies that are deploying virtualization technologies are also able to re-deploy freed-up IT resources to other projects rather then lay them off. (See Aberdeen's March 2011 report on *Storage Virtualization: Experience Begets Benefits* for more insight about how virtualization is changing job functions within IT). Virtualization should not be viewed by IT employees as a threat to their jobs, but as a tool to make their efforts more productive.

The second and third most widely cited benefits from desktop virtualization are the increased control IT gains over end-user desktops and over applications. This increased control can range from simply standardizing the number and versions of end-user applications to locking down end-user devices against any change, depending on the virtualization technology deployed. This greater control leads to reduced desktop downtime and faster recovery as IT can deal with dozens or hundreds of desktops from a central location.

Mobile users also see benefit from desktop virtualization. Depending on the type of desktop virtualization technology deployed, users can access their applications regardless of their location. For those that largely work offline, data can automatically synch whenever they reconnect. The company gains improved data security because no data is stored on the endpoint device, as server based software keeps everything on the central server. That way if the device is lost or stolen, no company data is accessible to non-approved individuals.

Desktop Performance Metrics

The following seven metrics were created from the information reported by organizations responding to the May 2011 Aberdeen desktop virtualization survey: change in time for routine desktop maintenance, change in desktop support spending over the last 12 months, number of desktop downtime incidents, change in desktop downtime over the last 12 months, the average time to remediate each downtime event, the amount of data lost in each downtime event, employee satisfaction with their desktops. Responding organizations were divided into two analysis groups: organizations that have had a desktop virtualization program for more than 1 year (labeled "with virtualization") versus those that have only just begun or have not yet implemented a program (labeled "no virtualization"). For the virtualized group, Aberdeen collected data on the desktop performance for both their virtualized and non-virtualized desktops.

These metrics are presented to allow readers of this report to gauge their own desktop program performance against this group of peers. For suggestions on how to improve your own program, the final section of this report offers recommendations for improving the desktop performance.

Table 4: Change in Time for Routine Maintenance over last 12 Months?

	With Virtualization	No Virtualization
Virtualized Desktops	- 20.4%	-
Non-virtualized Desktops	- 1.1%	- 2.7%

Source: Aberdeen Group, May 2011

This question was asked to learn how the amount of work supporting virtualized desktops has changed versus that for supporting non-virtualized desktops. Remember that on average companies have only deployed virtualization on 31% of their desktops. This 20.4% savings in reduced desktop support will become quite a substantial value when, in the future, it is applied across most or all desktops in the enterprise.

Table 5: Change in Desktop Support Spending Over the Last 12 Months?

	With Virtualization	No Virtualization
Virtualized Desktops	- 11.7%	-
Non-virtualized Desktops	- 0.2%	- 1.7%

Source: Aberdeen Group, May 2011

This is a key metric for understanding the ROI of a virtualized desktop program. This metric speaks to the amount of savings that can be gained from improving operational efficiencies and subsequently reducing the number of desktop support personnel. A 12% savings is not very large in absolute terms but will become larger as more than the 31% of current desktops get virtualized.

Comparing this metric against the previous one, we see that even though the amount of reported maintenance for desktop virtualized is reduced by around 20%, the reduced spending is only 12%. Desktop support resources are not eliminated but redeployed to other areas of IT, such as supporting the non-virtualized desktops.

Table 6: Desktop Downtime

	With Virtualization	No Virtualization
How many downtime incidents on average per year per knowledge worker?		
Virtualized Desktops	2.9	-
Non-virtualized Desktops	6.8	6.2
Change in desktop downtime over the last 12 months?		
Virtualized Desktops	- 23.4%	-
Non-virtualized Desktops	- 3.0%	- 6.8%

Source: Aberdeen Group, May 2011

Telephone: 617 854 5200
Fax: 617 723 7897

The next two metrics are interconnected. The number of reported downtime instances for virtualized desktops are half that of non-virtualized machines. Also, the amount of overall desktop downtime has been reduced by 23.4% for virtualized desktops. These metrics reflect several factors revealed in Aberdeen's data. First is the fact that 45% of companies that have deployed desktop virtualization report using thin or stateless clients (Table 13). These devices have no hard drives or other components that are commonly prone to failure. Secondly, the most common form of desktop virtualization deployed by respondents to this survey is server-based application hosting. Today's servers have many high availability features that will provide superior uptime to that provided by stand-alone PCs. Desktops that depend on thin clients with applications running on servers will have vastly superior uptime performance than standard PCs.

Table 7: Recovery and Data Loss

	With Virtualization	No Virtualization
Average time to remediate each downtime event		
Virtualized Desktops	1.0 hours	-
Non-virtualized Desktops	4.3 hours	3.0 hours
How much data is lost per downtime event in terms of hours of work		
Virtualized Desktops	.9 hours	-
Non-virtualized Desktops	2.3 hours	2.4 hours

Source: Aberdeen Group, May 2011

Recovery is faster and the amount of data lost is reduced when applications run on a server and have their centralized data backed up as part of the datacenter backup and recovery process. In desktop virtualization, patches for the Operating System (OS) and desktop applications are handled centrally, allowing for a large number of PCs to be updated with a single fix. Furthermore, site recovery services allow data centers to recover from downtime faster - with desktop virtualization, those performance benefits are passed on to the individual users in the form of faster recovery and higher application uptime.

Table 8: Employee Satisfaction (10 is the highest, 1 lowest)?

	With Virtualization	No Virtualization
Virtualized Desktops	7.2	-
Non-virtualized Desktops	6.9	6.0

Source: Aberdeen Group, May 2011

To consider a desktop virtualization program successful, the end-users must like, or at least accept, the changes and performance of their desktops. The process should not add any additional steps to end-user administration, be easy to use, and provide support for rich data and experience.

One interesting note is that for six of the previous seven metrics, the desktop performance of the "no virtualization" companies is superior to those of the non-virtualized desktops for "with virtualization" companies. One likely explanation, according to Aberdeen's data, is that companies that have deployed desktop virtualization have done so to the easier, least challenging use-cases, such as small office desktops, or those of task-based users. Managing the remaining, more challenging desktops results in lower performance metrics for their non-virtualized machines. The metric in Table 8 is the only metric where the non-virtualized desktops of companies in the "with virtualization" category outperform all others. Aberdeen believes this is an effect of the virtualization program. Even end-users that have not gained the advantages offered by virtualization understand that their company is addressing their desktop issues. Eventually, they know their company will get to them, so they are feeling some of the corporate attention addressed to improving desktop performance (Table 10 shows the benefits of measuring end-user satisfaction).

Aberdeen makes a point to look at technologies holistically. Generally a new technology will not succeed unless it is supported with organizational processes, capabilities, and other technologies that support the overall goal of the solution in question. The following section will discuss the most widely deployed capabilities and enablers as reported by our survey respondents.

What Companies are Doing to Gain Superior Results

Desktop virtualization, more so than virtualization for servers or storage, requires an upfront organizational investment. When server virtualization is deployed, success is achieved if end users can see no difference in the look or feel of an application. Desktop virtualization directly and visibly touches every user in the program. These upfront corporate investments include those shown in Table 9.

Table 9: Desktop Capabilities

	With Virtualization	No Virtualization
Standardized role and user group definitions	60%	11%
Formal education plan for IT in implementing desktop technology	55%	11%
Formal, documented desktop policies	54%	7%
Executive visibility into project	45%	19%

Source: Aberdeen Group, May 2011

Sixty percent (60%) of companies with desktop virtualization programs report that they divide their end-user community into role or usage group profiles. By creating groups of like users, IT can select the desktop

Telephone: 617 854 5200
Fax: 617 723 7897

virtualization tool that best meets that group's connectivity and usage patterns. It also allows IT to make upgrades across the entire group with a single change.

Aberdeen's data on the suggestions of knowledgeable desktop virtualization professionals shows a strong recommendation for not deploying desktop virtualization to the "easiest" groups first. The most difficult user groups (i.e. those with mobility or high bandwidth requirements) or those that utilize compute intensive applications will likely be of greater help in identifying the desktop virtualization solution that is the best fit for the needs of the organization. Expert users also caution that standardizing end-user roles and defining desktop models could be the largest challenge of deploying desktop technology, as end-users may work to avoid being categorized into a template that places any limits on their current and perceived future requirements.

Another process that is highly utilized by leading organizations is to establish a formal training program for the IT personnel that will be required to implement or manage new technologies. Fifty-five percent (55%) of deploying organizations report having their IT staff formally trained in implementing desktop technologies. Most vendors offer formal training classes on their products and this should be part of the desktop project. Training needs to be on-going and provided in a just-in-time schedule for new employees or those assigned to these programs.

Forty-five percent (45%) of deploying organizations reported that they have senior executive visibility into their desktop program versus just 19% for non-virtualized companies. Time and again, Aberdeen surveys find that having a named senior manager in charge of a program is a best practice. A senior manager can keep focus on a project and ensure it stays top of mind to those charged with its execution. Senior level interest keeps the desktop virtualization program from becoming just another task on a long IT work lists.

Table 10: Desktop Measurements

	With Virtualization	No Virtualization
Real-time data collection and analysis of desktop performance	40%	7%
Measure end-user satisfaction with desktop services	43%	11%

Source: Aberdeen Group, May 2011

IT must always be proactive when addressing any negative events that could affect the performance of computing assets. IT cannot wait for an end-user to inform them that there is an issue with desktop performance. Instead, they must be proactively monitoring the performance of virtualized desktops so they are alerted immediately to any negative event. Forty percent (40%) of companies that have deployed desktop virtualization monitor desktop performance in real-time.

Unhappy users can quickly derail a desktop virtualization project. If they complain loudly enough to their management about the negative impact to their productivity, it could put an entire desktop virtualization program in jeopardy. Part of the process of pre-selling a desktop virtualization program is understanding the end-user issues with both the current and planned desktop environments. A formal end-user satisfaction measurement process will show, at the very least, that IT is interested in their opinions and are using them as part of the desktop program.

Table 11: Desktop Technologies

	With Virtualization	No Virtualization
Thin or stateless clients	45%	4%
Support for tablet PCs	40%	0%
Support for employee-owned devices	39%	8%

Source: Aberdeen Group, May 2011

Thin or stateless clients can be an attractive alternative to traditional desktop PCs when included in a virtualization program. They are easy to deploy as they have no disk drive and little to no software installed. Thin clients may consume as little as 15Watts of power, compared to 100-150Watts for a full feature desktop computer, contributing to energy savings for the company. Forty-five percent (45%) of companies deploying desktop virtualization include thin or stateless clients in their program.

The next two enablers are tied to the contentious question of whether to support employee-owned devices, including tablet PCs. Sixty percent (60%) of survey respondents reported that they currently support only corporately owned devices. However, according to overall trends across several IT surveys, Aberdeen sees a movement among the top performing organizations toward supporting employee-owned devices. The experience of these organizations is that employees will connect their own devices, such as tablet PCs and mobile phones, to the company network and expect them to work, no matter what the formal corporate policy is. Experienced IT professionals believe it is preferable to deal with employee-owned devices in the open and as part of corporate policy rather than IT having to enforce an unpopular company policy. Most desktop virtualization technologies have ways of including multiple employee devices without opening security holes or negatively affecting the corporate performance overall.

Telephone: 617 854 5200
Fax: 617 723 7897

Case Study

Among the organizations hardest hit by the recent economic downturn are local and city governments. They have faced drastic budget and headcount cuts but are expected to continue to provide services to their residents. City IT departments feel this pressure especially as computing devices age and need to be replaced.

Aberdeen interviewed an IT manager from a mid-sized city in central Texas to see how they were using desktop virtualization to address their end-user support issues. As they are a public institution they can not go on record endorsing a technology so they requested their names and location be kept confidential.

The city had several pressures working against them. In addition to drastic budget cuts and a reduction in their IT headcount, they had to upgrade to Windows 7, had older PCs not certified for this operating system and had no remote monitoring tools. With the headcount reduction, servicing several sites required remaining IT technicians to drive to the affected locations.

With hundreds of desktops to support in multiple locations, the city IT manager selected a desktop virtualization technology that could support centralized PC management, support for disconnected users and remote sites. They could not afford to purchase new PCs so older devices still needed to be supported.

"We have to be very careful how we spend taxpayer money. Our goal is not only to reduce our costs but gain efficiencies in management and city employee satisfaction," said the IT Manager in charge of this program. "We needed the older PCs to keep working as we had no money to purchase new ones."

The city chose a virtualized infrastructure solution so the Windows 7 upgrade was performed once at the server level. The older PCs became thin clients to the server. As additional PCs reach the end of their useful life, they too will be converted to thin clients or new zero client devices will be purchased.

"We are happy with our choice of a server-based desktop virtualization solution. We delayed having to invest in new PCs and enabled remote monitoring and PC management. The technology has definitely improved and enables more flexibility than when we first looked at this technology a few years ago." They also reported looking at supporting PCs from the Cloud but feel they have left that option open for a later project when the financial situation improves.

Key Takeaways

Desktop virtualization technology has been around for many years. However it is mostly the large enterprises that have had desktop

Aberdeen Group
A Harte-Hanks Company

virtualization programs for that length of time. The majority of both small and mid-sized organizations have only recently begun to deploy this technology, with most having just begun a program or are planning to implement one shortly.

This particular type of virtualization technology requires more of a corporate pre-investment than either server or storage virtualization. As potentially every desktop user will be affected, much pre-work is required to ensure they are given a voice in the project, feel a sense of ownership to make it work and will ultimately accept the final results of the project.

When proposing this project for corporate approval, Aberdeen's research shows that the ROI will be less about financial returns and more about operational efficiency. While server and storage virtualization have strong financial benefits such as the reduction in data center power and cooling, and a reduction in the number of storage or servers, desktop virtualization's returns are more operational. A strong desktop virtualization program will reduce the number and shorten the duration of downtime events, make deploying and upgrading applications faster, and increase the productivity of desktop users. These are valuable returns but must be presented up front in the decision making process so desktop virtualization success can be recognized for what it delivers.

For more information on this or other research topics, please visit www.aberdeen.com.

Related Research
High Availability for Virtualized Applications: Protecting Against Unplanned Downtime; June 2011
Managing Virtualized Applications: Optimizing Dynamic Infrastructures; April 2011
Storage Virtualization: Experience Begets Benefits; February 2011
The Case for Data Deduplication in Highly Virtualized Environments; November 2010
The Case for Virtual I/O Devices; October 2010
Why are IT Leaders Deploying Virtualized Storage?; June 2010
Author: Dick Csaplar, Senior Research Analyst, Virtualization and Storage Practice (dick.csaplar@aberdeen.com)